

Cybersecurity and National Defense

Research Project Proposals

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1 Cyberattack Case Studies

1.1 Stuxnet: Cyber-Physical Sabotage and Strategic Implications

Stuxnet was one of the first cyberattacks designed to cause physical damage to industrial equipment. It targeted Iran's nuclear facilities by secretly manipulating industrial control systems while hiding its actions. This research aims to understand what happened during the attack, how it was technically carried out, and why it was successful. It also examines the strategic implications of Stuxnet and how it changed global thinking about cyber warfare by showing that cyber operations can cause a national-level impact.

1.2 SolarWinds: Software Supply Chain Compromise

The SolarWinds attack involved malicious code being inserted into a trusted software update, allowing attackers to access many organizations. This research aims to understand how the compromise happened, how it spread through the software supply chain, and why it remained undetected for so long. It also explores the broader security risks linked to software supply chains and the challenges of protecting interconnected systems.

1.3 Colonial Pipeline: Ransomware and National Energy Security

In 2021, a ransomware attack forced the Colonial Pipeline to temporarily shut down fuel distribution. This research aims to understand how the ransomware attack occurred, how it affected operations, and why the company decided to halt services. It also examines the wider impact on national energy security and what the case reveals about protecting critical infrastructure from cybercrime.

1.4 WannaCry: Patch Management Failure at Global Scale

WannaCry was a ransomware attack that spread rapidly in 2017 by exploiting a known software vulnerability. This research aims to understand how the malware spread so quickly, why many systems were unpatched, and what damage it caused worldwide. It also analyzes the importance of patch management and basic cybersecurity practices in preventing large-scale disruption.

1.5 NotPetya: Destructive Malware Disguised as Ransomware

NotPetya appeared to be ransomware but was actually designed to destroy data. It began in Ukraine and quickly spread globally, causing massive financial losses. This research aims to understand how the attack unfolded, how it spread beyond its initial target, and why it was

so destructive. It also explores the blurred line between cybercrime and state-sponsored cyber operations.

1.6 Mirai: IoT Botnets and Internet Fragility

Mirai infected thousands of insecure Internet of Things (IoT) devices and used them to launch large Distributed Denial of Service (DDoS) attacks. This research aims to understand how the malware infected devices, how the botnet was created, and how it was used to disrupt online services. It also examines what this case reveals about weak IoT security and the fragility of Internet infrastructure.

1.7 MOVEit Mass Exploitation Campaign

The MOVEit campaign exploited a vulnerability in widely used file transfer software, leading to large-scale data theft. This research aims to understand how attackers discovered and exploited the vulnerability, how the attack spread across organizations, and what data was compromised. It also analyzes what this case shows about vulnerability management and third-party software risks.

1.8 Log4Shell: Vulnerability at Internet Scale

Log4Shell was a critical vulnerability in the widely used Log4j software library. This research aims to understand how the vulnerability worked, why it was so dangerous, and how quickly it was exploited. It also examines what the case reveals about systemic risk in open-source software and the difficulty of securing widely shared digital components.

1.9 Equifax Data Breach

The Equifax data breach exposed sensitive personal information due to an unpatched software vulnerability. This research aims to understand how the breach occurred, why the vulnerability was not fixed in time, and how attackers accessed sensitive data. It also explores the legal, financial, and reputational consequences of the breach and the importance of stronger cybersecurity governance.

2 Malware

2.1 Malware Evasion Mechanisms

Modern malware is designed not only to infect systems but also to avoid being detected. Many malicious programs use techniques such as encryption, obfuscation, sandbox detection, or code modification to hide from antivirus software and security researchers. This research aims to understand how malware avoids detection and why these evasion techniques are effective. It examines common strategies used by attackers and analyzes real examples of malware that successfully bypassed security tools. The goal is to better understand the challenges defenders face and how security systems can be improved to respond to these tactics.

2.2 Malware Analysis Methodologies and Their Practical Applications

Malware analysis focuses on examining malicious software in depth in order to understand how it operates and what damage it is designed to cause. Analysts study the internal structure of malware using techniques such as static analysis, where the code is inspected without execution, and dynamic analysis, where the sample is observed in a controlled environment. This research explores how these investigative methods are applied in practice to identify indicators of malicious code. It evaluates the strengths and limitations of different analytical approaches and considers their role in incident response and threat intelligence activities.

2.3 Malware Detection Strategies and Their Operational Deployment

Malware detection is concerned with identifying threats within active systems and networks before they can cause significant harm. Organizations rely on a range of defensive mechanisms, including signature-based tools, behavioral monitoring systems, and machine learning models integrated into security platforms. This research examines how malware detection strategies work and how they are deployed in real operational environments, such as corporate or government networks. It can analyze issues such as scalability, false positives, performance constraints, and long-term maintainability, with the aim of understanding what makes a detection system both effective and sustainable in everyday cybersecurity practice.

2.4 The Evolution of Malware: From Early Self-Replicating Viruses to Advanced Persistent Threats (APTs)

Malware has changed significantly over time. Early computer viruses were often simple programs created to spread and cause disruption. Today, some malware is highly sophisticated and linked to long-term campaigns known as Advanced Persistent Threats (APTs), often associated with state actors. This research aims to understand how malware has evolved over the years, both technically and strategically. It explores key milestones in malware development and examines how motivations have shifted from curiosity and disruption to espionage, financial gain, and geopolitical competition. The study helps explain how the threat landscape has become more complex.

2.5 Mobile Malware and Smartphone Security

Smartphones store large amounts of personal and financial information, making them attractive targets for attackers. Mobile malware can steal data, spy on users, or send premium messages without consent. This research describes how mobile malware infects devices and what makes smartphones vulnerable. It can also examine how mobile operating systems attempt to protect users and what security practices can reduce risk.

3 Privacy, Data Protection, Surveillance

3.1 Pegasus Spyware: Surveillance and Sovereignty

This research begins with an explanation of what spyware is and how tools like Pegasus infect smartphones, gain administrator privileges, and extract data such as messages, calls, location, and microphone recordings without the user's knowledge. This research explains, in simple

terms, what makes this software powerful and difficult to detect. The second part analyzes its role in national defense and intelligence operations. The research may also discuss the use of cyber espionage tools to protect national security, and when their use undermines trust, sovereignty, or democracy.

3.2 Social Media Data Exploitation and Information Warfare

This research begins by explaining how social media platforms collect data from their users. Many apps gather information such as likes, shares, comments, location data, and even personality quiz results. This practice is often called “data harvesting”. It means collecting large amounts of personal information, sometimes without users fully understanding how their data will be used. A well-known example is the Cambridge Analytica case. The next step of the research is to explain what data harvesting and microtargeting mean in practice, examining how such targeted advertising can influence elections and public debate, often without public visibility or transparency. Finally, the research can evaluate whether existing data protection laws, such as the GDPR, are strong enough to prevent this kind of manipulation and to safeguard democratic processes.

3.3 Cryptographic Backdoors: Security vs. Lawful Access

This research explains how encryption protects digital communications by converting readable information into coded text using mathematical keys. It clarifies how technologies such as symmetric and asymmetric encryption, HTTPS, and end-to-end encryption secure everyday activities like messaging and online banking. The research then explores the debate over whether governments should have access to encrypted communications through “backdoors”, which would allow authorities to access encrypted data. The topic examines the tension between cybersecurity and privacy right, requiring strong encryption, and intelligence community’s needs to include exceptional access to user communications.

3.4 Data Breaches in Defense and Government Systems

This research examines how data breaches occur in corporate, defense and government systems, focusing on technical causes such as weak authentication, phishing attacks, insider threats, software vulnerabilities, and misconfigured cloud services. It can refer to real-world cases such as the Office of Personnel Management breach (2015), the SolarWinds supply chain attack (2020), and the 2023 U.S. Department of Defense email exposure to illustrate how sensitive information can be compromised. The study then analyzes the consequences of such breaches, e.g., highlighting how the loss of classified or strategic data can undermine national security and defense capabilities.

4 Artificial Intelligence (AI) and Cybersecurity

4.1 AI in Cyber Defense and Cyber Offense

This topic introduces how AI systems analyze large amounts of data to detect unusual patterns or predict threats. Students explain how AI can strengthen cyber defense by identifying attacks quickly. The research also considers how AI may be used offensively, for example to automate cyberattacks or disinformation campaigns. The discussion can also address strategic and ethical implications for national defense.

4.2 The Role of AI in Malware Development and Detection

AI is increasingly used in cybersecurity, but it can also be used by attackers. AI can help automate attacks or make malware more adaptive. This research aims to understand how AI is used both to create more advanced malware and to improve detection systems. It explores the risks and benefits of AI in cybersecurity and considers whether AI will change the balance between attackers and defenders.

4.3 Security and Privacy Attacks on Machine Learning Models

This research investigates how machine learning systems themselves become targets of cyberattacks. The study should classify major attack types, such as adversarial examples, data poisoning, model inversion, and membership inference attacks. It should explain how these attacks exploit vulnerabilities in training data, model architecture, or deployment environments. The discussion may extend to strategic risks in national defense contexts, especially when AI systems are deployed in intelligence analysis, autonomous systems, or critical infrastructure protection. Questions about accountability, resilience, and the protection of sensitive data can also be addressed.

4.4 AI-Based Anomaly Detection in Network Security

This research explores the application of artificial intelligence to detect abnormal behavior in network traffic. The study should first explain traditional rule-based intrusion detection systems and their limitations. It should then analyze how machine learning methods (e.g., clustering algorithms, autoencoders, recurrent neural networks) identify deviations from normal traffic patterns. The discussion may extend to national defense scenarios, such as monitoring military networks, protecting critical infrastructure, and detecting advanced persistent threats.

5 Critical Infrastructure and Cyber-Physical Systems Security

5.1 Critical Infrastructure and Cybersecurity: Protecting National Assets

This research describes the technical structure of critical infrastructure systems, such as power grids, water systems, communication networks. The research explains how these systems can be vulnerable to cyberattacks, including ransomware and denial-of-service attacks. The second part analyzes why protecting critical infrastructure is central to national defense and what strategies governments can use to strengthen resilience.

5.2 Cybersecurity in Smart Grids

Smart grids combine traditional power systems with digital communication networks, smart meters, and automated control. This research can analyze how data flows between control centers, substations, and field devices, and where attackers could interfere. A useful case study is the 2015 cyber attack on the Ukrainian power grid, which caused a large blackout affecting hundreds of thousands of people. This incident showed how attackers can move from IT networks into operational systems. From a national defense perspective, power grids are strategic

assets. A coordinated attack could disrupt military bases, emergency services, and communications. The research can examine how smart grid protection supports national resilience and crisis response.

5.3 Security of Industrial Control Systems (ICS/SCADA)

Industrial Control Systems, including SCADA systems, are used to manage power plants, water treatment facilities, factories, and pipelines. Research should focus on understanding how these systems operate, how legacy technologies increase risk, and how attackers may exploit remote access or outdated software. Beyond the technical level, these systems directly control physical processes. A cyber attack can cause real-world damage, environmental harm, or even loss of life. Key case studies of attacks to ICSs are Stuxnet and Colonial Pipeline. The research can also examine the impact of such attacks on national security, economic stability, and public trust.

5.4 Cybersecurity in Automotive Systems

Modern vehicles rely on complex electronic systems, internal networks, wireless connections, and software updates. The research can examine how in-vehicle networks such as CAN bus operate, how electronic control units communicate, and how external interfaces like Bluetooth, Wi-Fi, GPS, and over-the-air updates may introduce vulnerabilities. A well-known case study is the 2015 remote hack of a Jeep Cherokee, where security researchers were able to control steering and braking functions through a vulnerability in the infotainment system. This incident demonstrated that cyber attacks on vehicles can directly affect physical safety. From a national defense and public security perspective, large-scale exploitation of automotive systems could disrupt transport networks, emergency response, or military logistics. Research in this area can therefore connect technical vehicle security with road safety, supply chain protection, and national resilience in the face of coordinated cyber threats.

5.5 Cybersecurity in Aviation Systems

Aviation systems include aircraft communication links, airport management systems, and air traffic control networks. The research can explore vulnerabilities in communication protocols, satellite navigation systems, and ground infrastructure. Research may focus on secure communication standards, redundancy mechanisms, and intrusion detection for aviation environments. In 2015, a cybersecurity researcher demonstrated potential weaknesses in aircraft communication systems, raising global attention to aviation cybersecurity. In addition, various airports have suffered ransomware attacks in recent years, affecting operations and passenger services. From a defense and strategic point of view, aviation is critical for both civilian mobility and military operations. Disruption of aviation systems can affect national economy, emergency response, and defense logistics. This research topic can analyze how cybersecurity supports airspace safety, national sovereignty, and crisis management.

6 National Defense

6.1 Digital Sovereignty in the EU

Digital sovereignty has emerged as a strategic priority for European policymakers concerned with technological dependence and security vulnerabilities. This research explores how control

over data infrastructures, cloud ecosystems, semiconductor supply chains, and cybersecurity standards shapes national and regional autonomy. It evaluates whether current European initiatives effectively enhance strategic resilience or risk fragmenting the global digital landscape.

6.2 Quantum Threat to National Security

Advances in quantum computing pose a long-term challenge to current cryptographic systems that underpin military communications, intelligence operations, and critical infrastructure. This study examines the timeline and plausibility of cryptographically relevant quantum capabilities, as well as national preparedness for post-quantum migration.

6.3 DDoS-for-Hire & Critical Services: From Crime to Geopolitical Pressure

Distributed Denial-of-Service (DDoS) capabilities are increasingly commoditized through illicit “booter” services, lowering the barrier to disruptive cyber operations. This research explores how tools originally used for criminal purposes can also be used for political or strategic goals. In particular, it examines how DDoS attacks work, how they can shift from ordinary cybercrime to instruments of geopolitical coercion, especially when targeting critical services such as government systems, financial institutions, or infrastructure. The study can also analyze the increasingly unclear boundary between cybercrime and actions that may be connected to state interests. Finally, it discusses what types of defensive strategies are necessary to protect essential national services from these kinds of attacks.

6.4 Protection of Space-Based Assets from Cyber Threats

Satellites and other space-based systems play an essential role in everyday life. They support communication, navigation (such as GPS), weather forecasting, banking systems, and military operations. As these systems become more connected to digital networks, they also become vulnerable to cyber attacks. This research examines how space-based assets can be targeted through cyber means and what risks these attacks create for national security and civilian life. It explores possible weaknesses in satellite systems, ground control stations, and communication links. The study also analyzes what technical, organizational, and policy measures can be used to protect these critical systems. The goal is to better understand how to strengthen the cybersecurity of space infrastructure in an increasingly contested environment.

6.5 Supply Chain Attacks as Strategic Weapons

A supply chain attack happens when attackers target a trusted supplier, such as a software company or hardware provider, in order to indirectly compromise many other organizations. Instead of attacking a target directly, they exploit weaknesses in the production or distribution process. This research explores how supply chain attacks can be used not only for criminal purposes but also as strategic tools by states or politically motivated groups. The study also looks at the challenges of detecting and preventing such attacks, and what governments and companies can do to reduce the risks. The aim is to understand why supply chain attacks are becoming powerful instruments in modern geopolitical competition.

6.6 Cyber Espionage and State-Sponsored Hacking

This research examines how cyber espionage works from a technical perspective, explaining methods such as phishing, malware deployment, and the exploitation of software vulnerabilities, often organized in long-term operations known as Advanced Persistent Threats (APTs). It refers to historical cases such as Stuxnet, Operation Aurora, and the SolarWinds attack to illustrate how states use cyber operations for intelligence gathering, sabotage, or strategic advantage. The study then analyzes whether cyber espionage is becoming a regular tool of national defense and discusses how international law addresses issues such as sovereignty, attribution, and state responsibility in cyberspace.

7 Social Engineering and Security Awareness

7.1 Cognitive Biases as a Strategic Vulnerability in Cyber Operations

Social engineering attacks often succeed not because of technical weaknesses, but because of human behavior. This research explores how cognitive biases such as trust in authority, urgency, fear of loss, and overconfidence influence decision-making. Starting from the dual-process theory (intuitive vs. analytical thinking), the study analyzes how attackers exploit automatic reactions to bypass rational evaluation. The study can evaluate why even trained individuals can fall victim to manipulation, and how awareness programs can reduce this psychological vulnerability in corporate and national security contexts.

7.2 Phishing Evolution: From Mass Campaigns to Targeted Strategic Attacks

Phishing has evolved from generic email scams to highly targeted and strategic operations such as spear phishing, whaling, and Business Email Compromise (BEC). This research examines how phishing techniques have changed over time, focusing on personalization, social media intelligence (OSINT/SOCMINT), and supply chain targeting (i.e., attacks that exploit trusted business partners to reach the final victim). The study analyzes why targeted phishing is more effective and how it can impact critical organizations and government institutions. The study can also evaluate which defensive strategies are most effective in reducing organizational risk.

7.3 Ransomware and Social Engineering: The Human Factor Behind Initial Access

Many ransomware attacks begin with social engineering, often through phishing emails or malicious attachments. This research investigates how attackers gain initial access to networks by manipulating employees rather than exploiting technical vulnerabilities. It analyzes common entry points such as malicious code hidden inside documents, fake login pages, and credential harvesting. The study can also examine why awareness training sometimes fails and proposes practical measures to strengthen user resilience in critical infrastructure and public sector environments.

7.4 Security Awareness Programs: Do They Really Work?

Organizations invest significant resources in cybersecurity awareness training, but the real effectiveness of these programs is often unclear. This research evaluates different approaches to

security awareness, including simulated phishing campaigns, gamification, and behavioral-based training. It examines measurable indicators such as click rates, reporting rates, and long-term behavioral change. The study can discuss whether awareness programs truly reduce risk or simply create temporary improvements.

7.5 Insider Threats and Social Engineering within Organizations

Not all threats come from outside an organization. Insider threats may involve employees who are manipulated, coerced, or intentionally collaborate with attackers. This research explores how social engineering techniques are used to recruit or exploit insiders. It analyzes psychological, organizational, and structural factors that increase insider risk, such as dissatisfaction, financial pressure, or excessive access privileges. The study can discuss how governance, monitoring, and organizational procedures can reduce insider-related vulnerabilities in corporate and national defense environments.

8 Software, Supply Chain, and IoT

8.1 Smart Devices, IoT, and Security Vulnerabilities

This research begins by explaining what the Internet of Things (IoT) is and how connected devices communicate through networks. The research analyzes why many IoT devices lack strong security protections and how they can be exploited in cyberattacks. The second part considers how vulnerabilities in civilian technologies may create broader national security risks.

8.2 Fuzzing Techniques and Their Role in Cybersecurity

This research examines fuzzing as a general software testing and vulnerability discovery technique used in cybersecurity. The study should begin by explaining the fundamental concept of fuzzing: automatically providing large volumes of unexpected, random, or malformed inputs to a system in order to identify crashes, abnormal behavior, or security flaws. The research can classify the main categories of fuzzing, including black-box, grey-box, and white-box approaches, as well as mutation-based and generation-based methods. The study can also explore the types of vulnerabilities commonly discovered through fuzzing, such as buffer overflows, memory leaks, logic errors, and improper input validation. Students may examine how fuzzing is applied to different targets, including applications, network protocols, web services, and embedded systems. The discussion can be extended to the strategic importance of fuzzing in cybersecurity and national defense. The research may include the role of fuzzing in proactive vulnerability discovery, and secure software development lifecycles.

8.3 Software Bill of Materials (SBOM) and Software Transparency

Modern software is built from many smaller components, often developed by different companies or open-source communities. A Software Bill of Materials (SBOM) is a list that shows all these components. This research explains what an SBOM is and why it has become important after major supply chain attacks. It examines how an SBOM can help organizations quickly understand if they are exposed to a known vulnerability. The study also discusses practical limits, such as keeping the list updated and ensuring accuracy. From a national defense point of view, software transparency can reduce hidden risks in critical systems.

8.4 Hardware Supply Chain Security and Trusted Components

Cybersecurity is not only about software. Hardware components such as chips, routers, and sensors are often produced in complex global supply chains. This research explains how hardware can be modified, tampered with, or counterfeited before it reaches the final user. It explores risks such as hidden backdoors or malicious firmware. The research can also discuss why trust in physical components is essential for national security.

8.5 Dependency Confusion and Package Repository Attacks

Modern software developers often download external libraries from public repositories. Attackers can exploit this process by uploading malicious packages with similar names, hoping developers will install them by mistake. This type of attack is known as dependency confusion. This research explains how package management systems work and how attackers exploit trust in public repositories. It can also study prevention strategies, such as repository control policies and verification processes.

8.6 IoT Botnets and Critical Infrastructure Disruption

IoT devices such as cameras, routers, and smart home systems are often poorly protected. When many insecure devices are infected, they can form a botnet controlled by attackers. This research explains how IoT botnets are created and how they are used to launch large denial-of-service attacks. It examines how such attacks could target critical infrastructure like energy systems or government services. The study also looks at basic security improvements that could reduce these risks.

8.7 Secure Development Lifecycle (SDL) in Defense Software Projects

Software used in defense and critical infrastructure must follow strict development processes. A Secure Development Lifecycle (SDL) integrates security checks at every stage of software creation, from design to deployment. This research explains how SDL works in practice and why it is different from traditional development models. It analyzes tools such as code review, automated testing, and vulnerability scanning. The study can also discuss whether secure development standards are enough to prevent future supply chain attacks.

9 Cryptography

9.1 Post-Quantum Cryptography: Preparing for the Quantum Threat

Current public-key cryptographic systems, such as RSA and ECC, rely on mathematical problems that are difficult for classical computers. However, future quantum computers could break these systems using algorithms such as Shor's algorithm. This research examines what Post-Quantum Cryptography (PQC) is and why it is necessary. The research should explain how quantum computing threatens existing cryptographic standards and analyze the main families of PQC algorithms, such as lattice-based, code-based, and hash-based schemes. The study should also explore the practical challenges of migrating national infrastructure, and critical services to quantum-resistant solutions, including interoperability, performance, and long-term planning.

9.2 Lightweight Cryptography for Constrained and Embedded Systems

Many devices used in critical infrastructure, and IoT environments have limited processing power, memory, and battery capacity. Traditional cryptographic algorithms may be too heavy for these systems. This research focuses on lightweight cryptography, which aims to provide security with reduced computational cost. The research should explain why constrained devices require specialized algorithms and analyze examples of lightweight ciphers and authenticated encryption schemes. The study may also examine the balance between efficiency and security, and discuss how lightweight cryptography can support national resilience in environments such as smart grids, or industrial control systems.

9.3 Blockchain and Cryptography in National Security Contexts

Blockchain systems rely heavily on cryptographic mechanisms such as digital signatures, hash functions, and consensus protocols. While often associated with cryptocurrencies, blockchain technologies may also have applications in defense and secure data management. This research explores the cryptographic foundations of blockchain systems and evaluates their potential use in secure supply chains, identity management, and tamper-resistant records.

9.4 Homomorphic Encryption and Secure Data Processing

In many national security contexts, sensitive data must be processed without being exposed. Homomorphic encryption allows computations to be performed directly on encrypted data, producing encrypted results that can later be decrypted. This research introduces the concept of homomorphic encryption and explains how it differs from traditional encryption schemes. The research could analyze practical use cases, such as secure intelligence data analysis or privacy-preserving cloud computing. The study can also consider performance limitations and whether current solutions are realistic for large-scale defense applications.

10 Network and Cloud Computing Security

10.1 Zero Trust Architecture

Traditional network security models assume that systems inside a trusted perimeter are safe. However, modern threats show that attackers often operate from inside compromised networks. Zero Trust Architecture (ZTA) is based on the idea that no user or device should be trusted by default. This research explains the main principles of Zero Trust, such as continuous authentication, least privilege, and network segmentation. The research can analyze how Zero Trust can be implemented in corporate, government or military networks, and what operational challenges may arise. The study may also evaluate whether Zero Trust improves resilience against advanced persistent threats.

10.2 Intrusion Detection and Prevention Systems (IDS/IPS)

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) monitor network traffic to identify malicious activity. This research explains the difference between signature-based and anomaly-based detection. The research can analyze how IDS/IPS systems are deployed in real environments and what limitations they face, such as false positives and encrypted

traffic inspection. The study may also explore how these systems can support national defense by detecting espionage, sabotage, or coordinated cyber operations.

10.3 Network Encryption and Traffic Analysis

Encryption protects the content of communications, but metadata and traffic patterns may still reveal sensitive information. This research examines how protocols such as TLS and IPsec secure network communications. The research can also explore traffic analysis techniques and how adversaries may extract intelligence without decrypting content. The study can evaluate whether encryption alone is sufficient to protect military or government communications.

10.4 Cloud Computing Models and Data Exposure

Cloud computing changes the way organizations manage infrastructure and data. Security responsibilities are divided between the cloud provider and the customer. This research explains common cloud service models such as Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS). The study can analyze common misconfigurations and how they lead to data breaches. The research can also analyze real-world incidents and evaluate tools and governance practices that reduce configuration risk.

11 Trusted Computing and Trusted Execution Environments

11.1 Trusted Computing and Hardware Root of Trust

This research explains the concept of Trusted Computing and why modern systems require a hardware root of trust. It introduces components such as the Trusted Platform Module (TPM) and secure boot mechanisms, describing how they ensure that a system starts in a secure and verified state. The study examines how cryptographic measurements and remote attestation are used to detect tampering and how trust is established from hardware to software. The research can also discuss limitations, such as physical attacks, performance issues in constrained devices.

11.2 Trusted Execution Environments (TEEs): Architecture and Security Guarantees

This research focuses on Trusted Execution Environments and how they isolate sensitive code and data from the rest of the system. It explains what a TEE is, how it differs from traditional process isolation, and why it is useful in hostile environments. The study analyzes the security guarantees offered by TEEs and their practical applications, such as protecting cryptographic keys or biometric data. It can also examine known attacks against TEEs and discusses whether they can be considered fully trustworthy in high-security contexts.

11.3 Reputation Systems and Distributed Trust Models

In many distributed systems there is no central authority capable of guaranteeing the reliability of all participants. To address this challenge, some networks use reputation mechanisms, where nodes evaluate the behavior of other nodes and assign trust scores based on past interactions. These scores influence decisions such as whether to exchange information, accept transactions,

or cooperate with other nodes. This research could examine how reputation systems and trust propagation algorithms are used to improve reliability and security in decentralized environments. The study can explain how trust relationships can be represented as graphs of interacting nodes and how reputation scores are aggregated and propagated across the network. Possible examples include peer-to-peer networks (e.g. BitTorrent, Gnutella, eMule), blockchains and decentralized networks (e.g. Ethereum), or algorithms such as EigenTrust. The research could also analyze common vulnerabilities such as Sybil attacks or reputation manipulation. The research may also discuss whether alternative trust mechanisms could improve robustness or reliability. In this part, the study could present a new conceptual model or architecture and evaluate its potential advantages and limitations, compared to existing solutions.